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ZF - Division A

Slip Control Software Design

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REFERENCES

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1 ABSTRACT

This module determines the pressure command for Active Roll Mitigation ([ARM](#)). Furthermore the [ARM](#) engine controller is part of this module.

2 ACRONYMS

ARM Active Roll Mitigation

ROI Roll Over Index

SWA Steering Wheel Angle

AyMod 'Modified' Lateral Acceleration, that tries to predict the Lateral Acceleration

Prim Primary

Sec Secondary

WTP Wheel Transition Pressure

3 OVERVIEW

This document describes the purpose and behavior of Arm_Veh_Control.

3.1 DESIGN GOAL

[ARM](#) calculates the pressure command based on Roll Over Index ([ROI](#)). Both pressure request and [ROI](#) are used in the engine controller.

3.2 GENERAL REQUIREMENTS

1. [ARM](#) should prevent dynamic rollover.
2. [ARM](#) should prevent static rollover.
3. [ARM](#) should apply sufficient brake pressure to the selected wheels in order to reduce the kinetic energy.
4. [ARM](#) should be capable of requesting engine torque reduction.

3.3 DETAILED REQUIREMENTS

Requirement ID	
Location	

3.4 SYSTEM MODE REQUIREMENTS

This module needs only to be processed, when **ARM** is allowed.

3.5 FLOW-CHARTS AND DIAGRAMS

Fig 3.1 shows the wheels selection based on the sign of Arm.State.Veh_Turn_Direction. In general the wheels on the outside of the turn are used to avoid roll over.

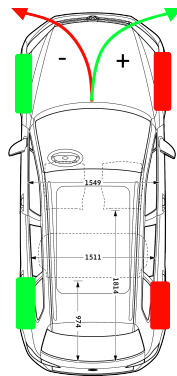


Figure 3.1: **ARM** Wheel Selection

4 DETAILED DESIGN

This chapter outlines the detailed design of Arm.State.

4.1 ALGORITHM DESCRIPTION

4.1.1 ArmVehicleControl

The function ArmVehControl selects the intervention wheels based on Arm.State.Veh_Turn_Direction (for further information see ArmState and Fig. 3.1).

In case Arm.Brake_Enabled is set and Arm is activated via Arm_Active flag, the roi_pressure_cmd is calculated based on **ROI**. The Trim VEH_PRESS_GAIN is one of the most important Arm trims, because the amount of Arm intervention can be tuned. In addition, the vehicle speed depending Speed.Gain is considered and allows a speed dependent modification of the VEH_PRESS_GAIN.

Since **ARM** pressure request is directly linked to **ROI**, its specific parameters influence directly the pressure request. Due to some calculations this behavior is not obvious. The

ROI_GAIN_TABLE, which is a function $f(\text{ay_mod_abs})$, is a crucial parameter. Without considering the Speed_Gain, it holds:

$$\text{roi_pressure_cmd} = \text{VEH_PRESS_GAIN} \cdot \text{ROI} \quad (4.1)$$

$$= \text{VEH_PRESS_GAIN} \cdot f(\text{ay_mod_abs}) \cdot (\text{ay_mod_abs} - \text{ay_mod_deact_thres}) \quad (4.2)$$

It is more convenient to change the ROI_GAIN_TABLE with respect to Ay_Mod with known deact_thres than with respect to ay_mod_abs. The Figure 4.1 below shows a suitable and an unsuitable parameter set.

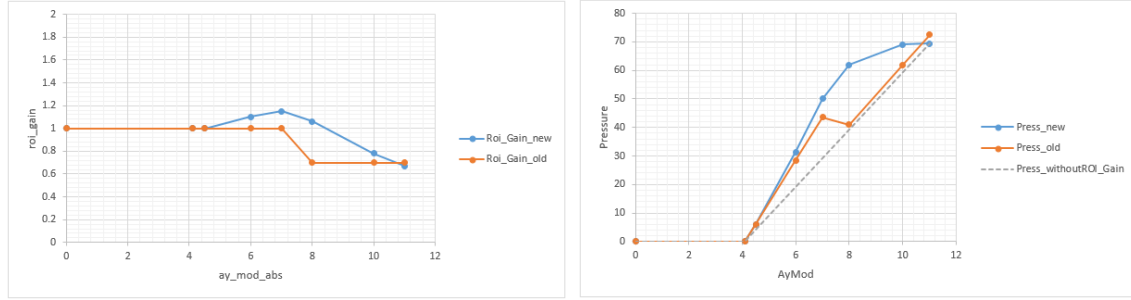


Figure 4.1: Influence of ROI on Pressure Command

Afterwards the function ArmRateLimitPressureCmd performs a rate limitation on the pressure command signal.

The ARM pressure request at the front axle is the result of the previous calculation. At the rear axle a certain amount of the front axle command is requested.

$$\text{FrontAxle: Pressure_Cmd_Prim} = \text{RateLimitation}(\text{Trim} \cdot f(\text{VehSpeed}) \cdot \text{ROI}) \quad (4.3)$$

$$\text{RearAxle: Pressure_Cmd_Sec} = \text{Trim} \cdot \text{Pressure_Cmd_Prim} \quad (4.4)$$

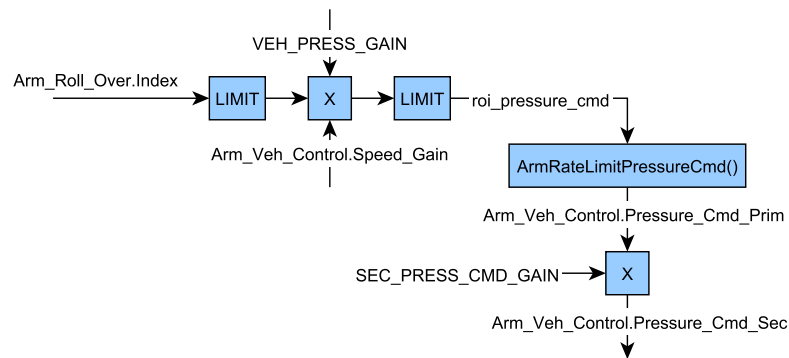


Figure 4.2: Signal Flow in ArmVehicleControl

4.1.2 ArmRateLimitPressureCmd

The function does not limit any increasing pressure command signal but only decreasing pressure commands are rate limited. It is possible to use different ramp down rates in case Arm_Active is set or not. Furthermore a timer observes the ramp down duration and shuts

down the ramp down, if the timer exceeds a trimable threshold.

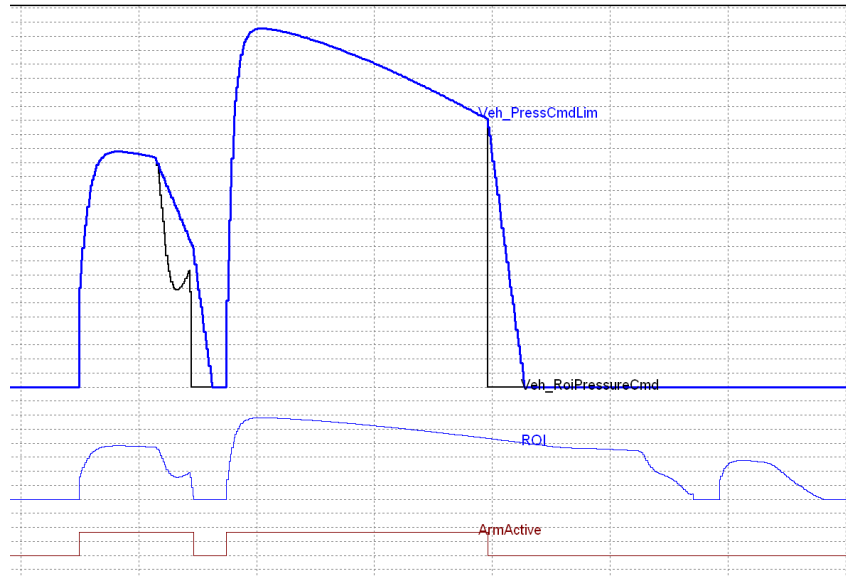


Figure 4.3: Rate Limiting

4.1.3 WtpFinitStateMachine

The **WTP** is designed for lane change maneuvers. Between the lanes, there is a short time with small lateral acceleration. Hence both front wheels have wheel load. The goal of **WTP** is to brake on both front wheels to reduce kinetic energy. Fig. 4.4 shows a lane change maneuver with **WTP**.

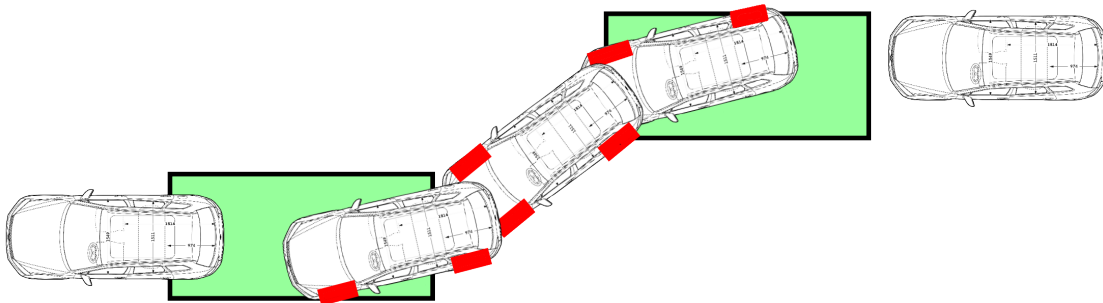


Figure 4.4: Lane Change with **WTP**

During lane change, **WTP** starts a prefill on the opposite **ARM** wheel. As soon as **Arm.State.Veh_Turn_Direct** changes, **WTP** holds the pressure request in the former **ARM** wheel. At this point, the car reduces speed and kinetic energy the most. The function is split into two subfunctions which are presented in the following sections.

WtpStateTransitions

Fig. 4.5 presents the **WTP** logic. It is implemented as finite state machine.

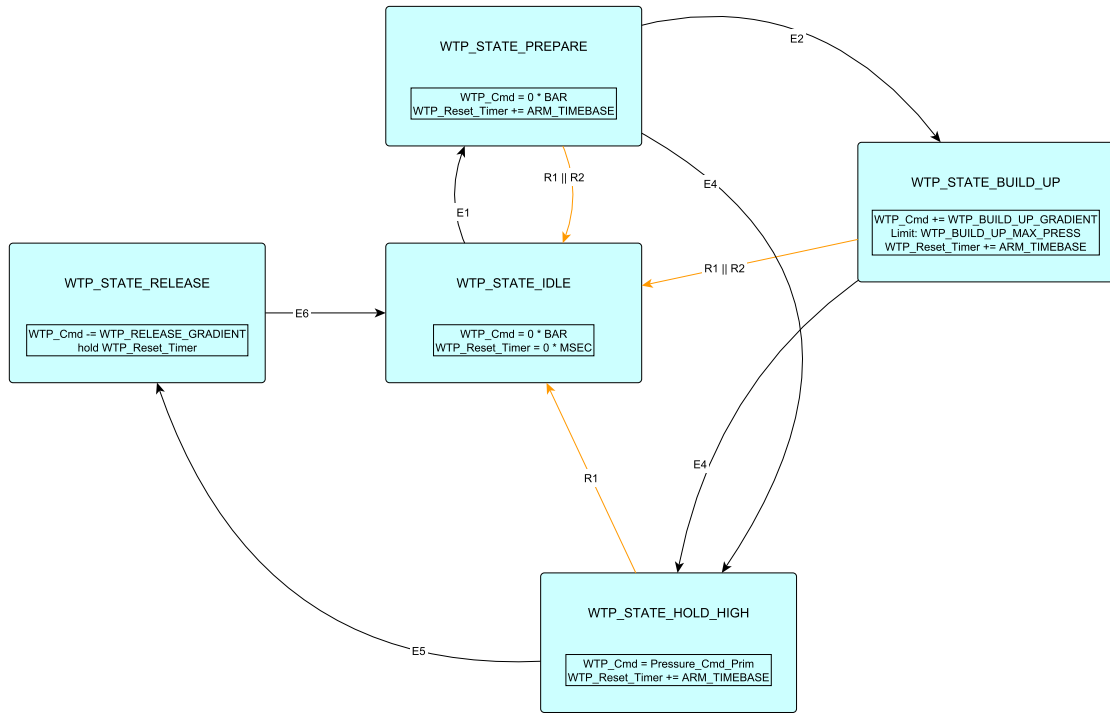


Figure 4.5: WTP Finite State Machine

with conditions:

C1:	$\text{Arm_Ay_Mod.Veh_Turn_Direction} \neq \text{Arm_Ay_Mod.Veh_Turn_Direction_Prev}$
C2:	$\text{Arm_State.Arm_Flag} = \text{TRUE}$
C3:	$\text{Arm_Input.Lat_Accel_Deriv} \cdot \text{Arm_Input.Lat_Accel} < 0$
C4:	$\text{ABS}(\text{Arm_State.Ay_From_Swa_Lead}) \geq \text{Arm_State.Ay_From_Swa_Lead_Thresh}$
C5:	$\text{Arm_State.Ay_From_Swa_Lead} \cdot \text{Arm_Input.Lat_Accel} < 0$
C6:	$\text{Arm_Veh_Control.Pressure_Cmd_Sec} \leq 0$
C7:	$\text{Arm_Veh_Control.WTP_Cmd} \geq \text{Arm_Veh_Control.Pressure_Cmd_Sec}$
C8:	$\text{Arm_State.Veh_Turn_Direction} \neq \text{Arm_State.Veh_Turn_Direction_Prev}$
C9:	$\text{ABS}(\text{Arm_Input.Lat_Accel}) < \text{WTP_LAT_ACCEL_THRESH}$
C10:	$\text{Arm_Veh_Control.WTP_Reset_Timer} \geq \text{WTP_RESET_TIMER_LIMIT}$
C11:	$\text{WTP_Cmd} = 0$

and corresponding event and reset transitions:

E1:	C1 AND C2
E2:	C2 AND C3 AND C4 AND C5
E3:	C6 OR C7
E4:	C4 AND C5 AND C8
E5:	C9
E6:	C11
R1:	C1 OR C10
R2:	C1 OR C10 OR (C8 AND NOT(C4 AND C5))

WtpStateActions

The actions during **WTP** are described in Fig. 4.5 as well. However in state **WTP_STATE_HOLD_HIGH**, it is possible to specify **WTP_HOLD_HIGH_MAX_PRESS** which is the pressure during hold phase. If the pressure request is too high when entering **WTP_STATE_HOLD_HIGH**, the pressure request is ramped down to **WTP_HOLD_HIGH_MAX_PRESS**. The gradient can be adjusted by setting **WTP_RELEASE_GRADIENT_0** to the desired values.

4.1.4 ArmEngineControl

The activation of **ARM** engine controller is based on **ROI** and is only possible, if **Arm_Engine_Enabled** is set. The engine control reduces a filtered drive torque dependent on the pressure command **Pressure_Cmd_Prim**. Afterwards **ARM** and **AyLimiter** engine commands are arbitrated and rate limited. Finally, the activation of **ARM** and **AyLimiter** engine controller is set.

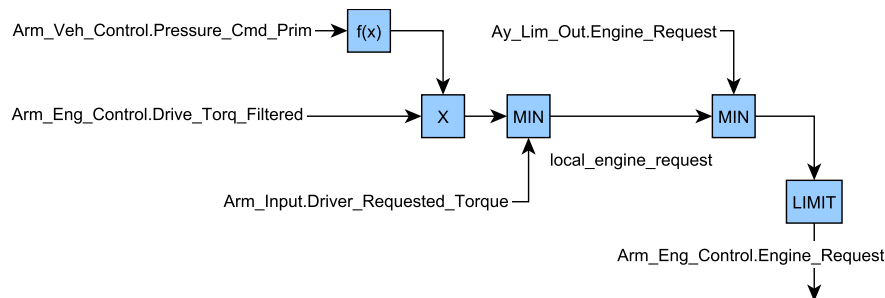


Figure 4.6: Signal Flow in ArmEngineControl

5 PROVIDED TASKS

5.1 ArmVehControl

Description	ArmVehControl
Rate	Standard Arm timebase 5 - 20ms
Dependency	
SYNTAX	void ArmVehControl (void)

6 PROVIDED SERVICES

6.0.1 Private Services

- **ArmRateLimitPressureCmd**
- **ArmEngineControl**

- [WtpFinitStateMachine](#)
- [WtpStateTransitions](#)
- [WtpStateActions](#)

7 CONFIGURATION OPTIONS

None.

8 CONFIGURATION PROCEDURES

None.

9 TEST CASES

Test category	
Requirements	
Test Case ID	

10 CALIBRATION INFORMATION

10.1 Pressure Request

After the activation, deactivation and signal processing steps are performed, the most crucial calibration steps need to be done. The optimization problem is to find a pressure that reduces as much energy as possible. However this pressure will yield a certain slip and therefore has an impact on steering and on tire wearing.

10.1.1 Pressure Request: Front Axle

The pressure request is derived from the roll over index. The most important parameters are

- **VEH_PRESS_GAIN**
- **PRESS_CMD_SPEED_TBL**

These parameters are used to amplify the roll over index and thus the primary pressure request.

Remark: Reduce the **VEH_PRESS_GAIN**, if the wheel slip is too high or the intervention is too harsh.

10.1.2 Pressure Request: Rear Axle

The pressure request for the rear axle is directly derived from the primary pressure request. There is a gain **SEC_PRESS_CMD_GAIN** that is used for scaling.

Remark: If the rear axle starts lifting off the ground there is less wheel load for reducing vehicle speed. Therefore a gain below 100% is suggested. However as long as the vehicle remains stable (no oversteer) the potential of the rear axle needs to be exploited.

10.1.3 Wheel Transition Pressure (WTP)

WTP can be deactivated by setting **WTP_RESET_TIMER_LIMIT** to zero. This will allow a state transition too the prepare state and an immediate reset. A suitable maneuver is a (double) lane change. In fact all **WTP** parameters are directly linked to the shape of the feed forward control. These parameters are outlined in Fig. 10.1

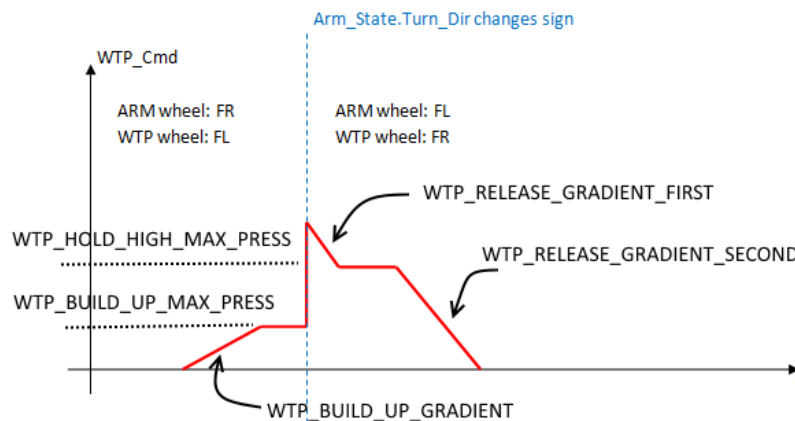


Figure 10.1: Shaping the **WTP** Request

10.1.4 Ramp Out

There is a rate limiting that affects the pressure ramp out. It is advised to start with the core trimming. If the ramp out is too slow, one can adjust **PEAK_PRESS_RAMPOUT_TBL** for a general approach. In addition there is a second gain that amplifies the ramp out as soon as a stable vehicle state is achieved. It is called **RAMP_LIM_FAST_RAMP_GAIN**.

Remark: A fast ramp out can cause disturbed vehicle reactions and an unease of the driver. Whereas a slow ramp out can feel like an elastic strap that holds the vehicle back.

10.2 Engine Request

The engine request for **ARM** excluding AyLimiter is dominated by **ENGINE_TRQ_TBL**. One can reduce the engine torque if necessary according to the actual pressure request. It is advised to start with the core trimming.

11 APPENDIX